# VARIABLE PROJECTION FOR MULTIPLE FREQUENCY ESTIMATION

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## ABSTRACT

The estimation of the frequencies of multiple complex sinusoids in the presence of noise is required in many applications such as sonar, speech processing, communications, and power systems. According to previous work [1, 2], this problem can be reformulated as a separable nonlinear least squares problem (SNLLS). In this paper, such formulation is derived, and a variable projection (VP) optimization is proposed for solving the SNLLS problem and estimate the frequency parameters. We also apply a lethargy-type theorem for quantifying the difficulty of the optimization. Moreover, an alternative procedure that speeds up the computation of the exact gradient is presented. Simulation results reveal that the proposed algorithm outperforms existing methods in terms of the MSE.

***Index Terms*—** frequency estimation, separable nonlinear least squares problem, variable projection, lethargy theorem

## 1. INTRODUCTION

The constant growth of engineering applications involving frequency estimation has motivated the appearance of numerous algorithms over the past few decades. Among the most well-known estimation strategies are the multiple signal classification (MUSIC) [3], its extension Root-MUSIC [4], and the estimation of signal parameters via rotational invariance techniques (ESPRIT) [2]. These subspace methods exploit the spectral decomposition of the covariance matrix for achieving high resolution. Some of the limitations of these algorithms include a high computational complexity as a consequence of the required computation of the eigenvalue decomposition of the measurement signal's covariance matrix [5, 6], and a degradation of the performance in scenarios with low signal to noise ratios (SNRs). Other approaches proposed a peak search on the magnitude of the discrete-time Fourier transform (DTFT) of the signal. The main advantage of these kinds of algorithms is that they have a low computational cost, but in the presence of multiple frequencies the estimation becomes biased due to the interactions of the different frequencies [7]. Another limitation of the DTFT is that in scenarios with very closely spaced frequencies, it is unable to separate the peaks that correspond to the frequency components. Other algorithms have also been proposed in [7–12].

Golub and Pereyra stated in their work [1] that frequency estimation can be formulated as SNLLS fitting problems for linear combinations of complex exponentials, where the linear coefficients represent the (complex) amplitudes, while the nonlinear ones are the frequencies of the signals. Such representation encourages the use of VP-like algorithms, but as it is indicated in [2] no further investigations were done due to the costly computations involved. Singular value decomposition (SVD) methods are typically used in these algorithms which are well known for having a high computational complexity. These types of algorithms were considered too expensive until fairly recently [1]. However, the increasing power of modern computers has rendered some of those arguments [1], which motivated this work.

Based on the previous statements [1, 2] the presented study provides a formulation of the frequency estimation problem in the framework of $\Gamma$-approximation which allows the application of lethargy theorems for quantifying the difficulty of the optimization. Comparisons with other state-of-the-art methods are also provided and an alternative method that accelerates the computation of the exact gradient is introduced. However, we want to point out that the original formulation based on SVD is more accurate but computationally expensive. The new approach instead guarantees a reasonable complexity-accuracy tradeoff.

## 2. NUMERICAL ANALYSIS OF THE FREQUENCY ESTIMATION PROBLEM

The problem of multiple frequency estimation can be analyzed in the framework of $\Gamma$-approximation, which has avoided attention so far. Following the work of Jupp [13], let us consider a uniformly convex Banach space $E$ with twice continuously Frechet differentiable norm $\|\cdot\|_E$, and an $E$-valued function $\gamma(f;t)$ ($f \in [a,b] \subseteq \mathbb{R}$, $t \in \mathbb{R}$), which is twice (strongly) differentiable with respect to $f$. Then,

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$\gamma$-polynomials, of order $n$, can be defined as follows:

$$\sigma(\mathbf{c}, \mathbf{f}) \equiv \sigma(\mathbf{c}, \mathbf{f}; t) = \sum_{j=1}^{n} c_j \gamma(f_j; t) \qquad (t \in \mathbb{R}), \tag{1}$$

where $\mathbf{c} \in \mathbb{C}^n$, and $a < f_1 < f_2 < \ldots < f_n < b$ is a subdivision of the interval $(a, b)$ by $n$ distinct points. Frequency estimation can be considered as a special case in which $E = L_2$ is a Hilbert space, $\sigma(\mathbf{c}, \mathbf{f}; t)$ is the nonlinear model of a complex valued signal $y$, the coefficients $c_j$'s denote complex amplitudes, $f_j \in [0, 0.5]$ stand for the frequencies, and $\gamma(f_j; t) = \exp(i2\pi f_j t)$. In case of frequency estimation, the signal $y \in L_2$ is given, and the parameters $\mathbf{c}$ and $\mathbf{f}$ are to be estimated. This can be done by solving the following nonlinear optimization problem:

$$\underset{(\mathbf{c},\mathbf{f}) \in \mathbb{C}^n \times s_n}{\arg\min} \; F(\mathbf{c}, \mathbf{f}) = \underset{(\mathbf{c},\mathbf{f}) \in \mathbb{C}^n \times s_n}{\arg\min} \; \|y - \sigma(\mathbf{c}, \mathbf{f})\|_2^2, \tag{2}$$

where $s_n$ is an $n$-simplex that represents the parameter space:

$$s_n = \{\mathbf{f} \in \mathbb{R}^n : 0 < f_1 < f_2 < \ldots < f_n < 0.5\}.$$

Note that the functions $\gamma(f_j; t) = \exp(i2\pi f_j t)$ $(j = 1, \ldots, n)$ are linearly independent for any choice of $\mathbf{f} \in s_n$, therefore the "full" functional problem in Eq. (2) can be simplified to

$$\underset{\mathbf{f} \in s_n}{\arg\min} \; \widetilde{F}(\mathbf{f}) = \underset{\mathbf{f} \in s_n}{\arg\min} \; \|y - \sigma(\mathbf{c}(\mathbf{f}), \mathbf{f})\|_2^2, \tag{3}$$

where $\mathbf{c}(\mathbf{f})$ denotes the coefficient vector of the orthogonal projection of $y$ to the subspace $\mathrm{span}\,\{\gamma(f_j; t) : j = 1, \ldots, n\}$. In discrete case, $\mathbf{y} \in \mathbb{R}^M$ with $M$ samples, and thus $\mathbf{c}(\mathbf{f})$ is the least squares solution to Eq. (2) for a given frequency estimate $\mathbf{f}$. With these assumptions, the $\mathbf{f}$ values of the critical points of $F(\mathbf{c}, \mathbf{f})$ and $\widetilde{F}(\mathbf{f})$ are identical on $s_n$ (see e.g. [13, 14]). Note that the full and the reduced functionals can also be extended to the closure of $s_n$ including multiple confluent frequencies:

$$\overline{s}_n = \{\mathbf{f} \in \mathbb{R}^n : 0 \le f_1 \le f_2 \le \ldots \le f_n \le 0.5\}. \tag{4}$$

In case of complex exponentials (or sines and cosines for the real valued formulation), $\gamma(f_j; t)$, $\gamma'(f_j; t)$, and $\gamma(f_k; t)$ are linearly independent for any choice of $f_j, f_k \in [0, 0.5]$, therefore the critical points of $F(\mathbf{c}, \mathbf{f})$ and $\widetilde{F}(\mathbf{f})$ remain identical on $\overline{s}_n$ as well (see e.g. Lemma 2.1 and Theorem 5.1 in [13]).

The boundary hyperplanes of $\overline{s}_n$ with only one confluent frequency pair, i.e. $f_p = f_{p-1}$, are called the $p$th main-faces $s_n^{(p)}$, which are of particular importance to quantify the difficulty of solving Eq. (3). On each of these main-faces, the reduced functional $\widetilde{F}(\mathbf{f})$ is symmetrical with respect to exchanging the variables $f_{p-1}$ and $f_p$, which implies the following theorem.

**Theorem 1. (Jupp [13], "Lethargy Theorem")** *Across the main-faces $s_n^{(p)}$ $(p = 2, \ldots, n)$ of $\overline{s}_n$,*

$$\mathbf{n}_p^T \nabla \widetilde{F}(\mathbf{f}) = 0,$$

*where $\mathbf{n}_p$ is the unit outward normal to $s_n^{(p)}$.*

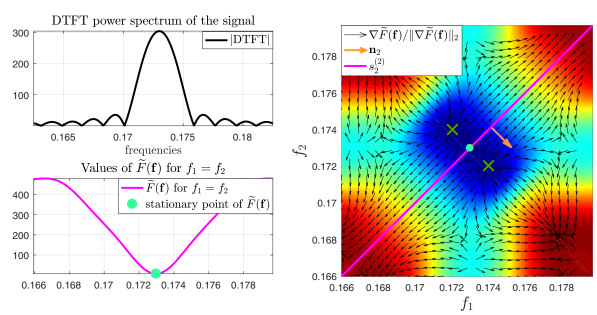
According to [15], the lethargy theorem has the following consequences:

(i) the reduced and so the full functional has many stationary points on the main-faces of $\overline{s}_n$;

(ii) $\widetilde{F}(\mathbf{f})$ and $F(\mathbf{c}, \mathbf{f})$ are non-convex for any set of data, and any choice of smooth convex norm;

(iii) numerical optimizers have poor convergence near the boundary of $s_n$.

The lethargy theorem explains why the problem of frequency estimation is a difficult task in general. Fig. 1(b) demonstrates this pheomena, where we displayed the values of $\widetilde{F}(\mathbf{f})$ and the corresponding normalized gradient vector field in 2D. Note that the outward directional component $\mathbf{n}_p$ of $\nabla \widetilde{F}(\mathbf{f})$ becomes small near the boundary, where $f_1 \approx f_2$. Therefore, numerical algorithms can be easily trapped near the main-faces $s_n^{(p)}$, which highlights the importance of good initialization. In our work, we chose the DTFT [5, 16, 17] estimate of the frequencies as initial points of the optimization and we assumed that the number of frequencies is known. In the top of Fig. 1(a), the single peak of the spectrum points out a frequency bin around which the cost function was evaluated. The symmetry of $\widetilde{F}(\mathbf{f})$ and its critical points can be seen in Fig. 1(b).




(a) Extreme points on $s_2^{(2)}$. (b) Graph of $\widetilde{F}(\mathbf{f})$ for $\mathbf{f} \in \mathbb{R}^2$.

**Fig. 1**: Values of $\widetilde{F}(\mathbf{f})$, where the signal contains two normalized frequency components $f_1 = 0.171$ and $f_2 = 0.174$.

## 3. EFFICIENT COMPUTATION OF THE VP GRADIENT

The minimization of the "reduced" functional in Eq. (3) leads to the solution of the "full" problem in Eq. (2). This fact was utilized by other authors to derive various frequency estimators like MUSIC [2]. We emphasize that, in former works, the optimization was restricted to subsequent estimations of single frequencies. On one hand, this approach eases the computational burden and allows simple implementations such as gridsearch. On the other end, Roy and Kailath [2] pointed out that these techniques have some drawbacks due to one-dimensional simplification, e.g. in low SNR scenarios, the well-known MUSIC algorithm has limited ability to resolve closely spaced frequencies. Therefore a simultaneous search for all the frequencies would be preferred, however, the optimization in multidimensional search space is more costly. In this section, we resolve this problem by speeding up the computation of the gradient of the reduced functional in Eq. (3).

From Eq. (2), we have that $\mathbf{c}(\mathbf{f}) = \mathbf{\Psi}^{\dagger}(\mathbf{f})\mathbf{y}$ is the minimal least squares solution for a fixed $\mathbf{f}$, where $\mathbf{\Psi}(\mathbf{f}) = [\mathbf{\Psi}_1(\mathbf{f}), \ldots, \mathbf{\Psi}_n(\mathbf{f})]$ denotes the matrix functions, $\mathbf{\Psi}_1(\mathbf{f}) = [1, e^{i2\pi f_1}, \ldots, e^{i2\pi f_1(M-1)}]$, and $\mathbf{\Psi}^{\dagger}(\mathbf{f})$ is the Moore–Penrose pseudoinverse. The linear operator $\mathbf{P}_{\mathbf{\Psi}(\mathbf{f})} = \mathbf{\Psi}(\mathbf{f})\mathbf{\Psi}^{\dagger}(\mathbf{f})$ is the orthogonal projector on the linear space spanned by the columns of the matrix $\mathbf{\Psi}(\mathbf{f})$ and $\mathbf{P}^{\perp}_{\mathbf{\Psi}(\mathbf{f})} = \mathbf{I} - \mathbf{P}_{\mathbf{\Psi}(\mathbf{f})}$ denotes the projector on the orthogonal complement of the column space of $\mathbf{\Psi}(\mathbf{f})$. Then considering $\mathbf{c}(\mathbf{f})$ in Eq. (3), the frequency parameters can be calculated by solving the following optimization

$$\underset{\mathbf{f}\in s_n}{\arg\min} \left\|\mathbf{y} - \mathbf{\Psi}(\mathbf{f})\mathbf{\Psi}^{\dagger}(\mathbf{f})\mathbf{y}\right\|_2^2 = \underset{\mathbf{f}\in s_n}{\arg\min} \left\|\mathbf{P}^{\perp}_{\mathbf{\Psi}(\mathbf{f})}\mathbf{y}\right\|_2^2 \quad (5)$$

where the resulting functional is a VP functional [14]. For simplicity, we omit the vector of free parameters $\mathbf{f}$ from the following notations. According to [18] the $k$th coordinate of the gradient of the functional is given by

$$\frac{1}{2}\nabla\widetilde{F}_k = \left[-(\mathbf{P}_{\mathbf{\Psi}}\mathbf{D}_k\mathbf{\Psi}^{\dagger} + (\mathbf{P}_{\mathbf{\Psi}}\mathbf{D}_k\mathbf{\Psi}^{\dagger})^T)\mathbf{y})\right]^T \mathbf{P}^{\perp}_{\mathbf{\Psi}}\mathbf{y}, \quad (6)$$

where $\mathbf{D}_k = \partial\mathbf{\Psi}(\mathbf{f})/\partial f_k$ represents the matrix of partial derivatives of $\mathbf{\Psi}(\mathbf{f})$ with respect to the single parameter $f_k$. For performing the VP optimization, we used the real representation of the problem (splitting into real $\mathcal{R}(\cdot)$ and imaginary $\mathcal{I}(\cdot)$ components). As a result of this transformation, the matrix functions, measurements vector, and the linear coefficients are given by

$$\tilde{\mathbf{\Psi}} = \begin{bmatrix} \mathcal{R}(\mathbf{\Psi}) & -\mathcal{I}(\mathbf{\Psi}) \\ \mathcal{I}(\mathbf{\Psi}) & \mathcal{R}(\mathbf{\Psi}) \end{bmatrix} \quad \tilde{\mathbf{y}} = \begin{bmatrix} \mathcal{R}(\mathbf{y}) \\ \mathcal{I}(\mathbf{y}) \end{bmatrix} \quad \tilde{\mathbf{c}} = \begin{bmatrix} \mathcal{R}(\mathbf{c}) \\ \mathcal{I}(\mathbf{c}) \end{bmatrix}.$$

One of the main reasons why VP methods have not been regarded in the past is because the simultaneous optimization of the frequencies is a computationally expensive task [2]. The computation of the gradient in (6) involves the calculation of the pseudoinverse $\mathbf{\Psi}^{\dagger}$ typically through an SVD method. SVD algorithms are well known for being costly. In order to ease the computational burden, we replace the SVD with a faster iterative calculation of the pseudoinverse (ICP) based on [7], the individual steps of ICP are described in Algorithm 1.

Fig. 4(a) illustrates number of operations over the number of measurements for SVD and ICP. Note that with the increase of the number of measurements the complexity of SVD

**Algorithm 1:** Iterative Computation of Pseudoinverse

**Input:** $\tilde{\mathbf{\Psi}}$
$[M, Q] = \text{size}(\tilde{\mathbf{\Psi}})$
$\mathbf{W}_Q = \frac{2}{M}\tilde{\mathbf{\Psi}}^T\tilde{\mathbf{\Psi}}$
$\mathbf{W}_{Q-1}^{-1} = 1$
**for** $k = 2 : Q$ **do**
  $\mathbf{x}_Q = \mathbf{W}_Q(1 : k-1, k)$
  $\delta_Q = \mathbf{x}_Q^T\mathbf{W}_{Q-1}^{-1}\mathbf{x}_Q$
  $\mathbf{0} = \text{zeros}(\text{length}(\mathbf{x}_Q), 1)$
  $\mathbf{y}_Q = \mathbf{W}_{Q-1}^{-1}\mathbf{x}_Q$
  $\mathbf{W}_{Q-1} = \begin{bmatrix} \mathbf{W}_{Q-1}^{-1} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix} + \frac{1}{1-\delta_Q}\begin{bmatrix} \mathbf{y}_Q\mathbf{y}_Q^T & -\mathbf{y}_Q \\ -\mathbf{y}_Q^H & 1 \end{bmatrix}$
**end**
$\tilde{\mathbf{\Psi}}^{\dagger} = \frac{1}{2M}\mathbf{W}_{Q-1}^{-1}\tilde{\mathbf{\Psi}}^T$
**Output:** $\tilde{\mathbf{\Psi}}^{\dagger}$

also grows considerably. The main reason why this happens is that the complexity of SVD scales by a power of two with the number of measurements $M$, while ICP scales linearly, see Table 1. The derivation of the complexity of ICP has been omitted due to page restrictions.

Although our approach is less time-consuming, we would like to clarify that SVD is the most stable and accurate way of computing the pseudoinverse. Nevertheless, ICP provides accurate results and no significant changes are observed in the final performance. In practical applications where computational time is a constraint, alternatives as ICP with a good complexity/accuracy tradeoff are necessary.

**Table 1**: Complexity

| Algorithm | operations | $\mathcal{O}(\cdot)$ |
|---|---|---|
| SVD | $4M^2 + 8Mn^2 + 9n^3$ [19] | $M^2$ |
| ICP | $(4M+1)n^2 - Mn + \frac{10n^3-15n^2-n+6}{6}$ | $Mn^2$ |

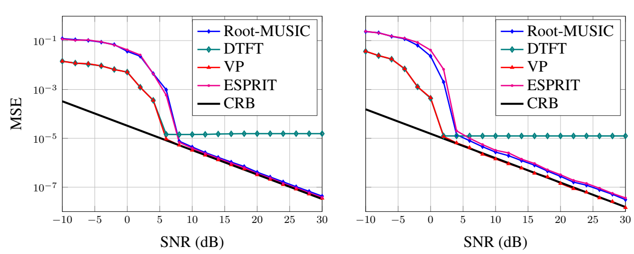
## 4. SIMULATION RESULTS

The simulation results are presented in this section. The performance is evaluated in terms of the mean square error (MSE)

$$\text{MSE} = \frac{1}{Rn}\sum_{r=1}^{R}\sum_{i=1}^{n}|\hat{f}_{i,r} - f_i|^2 \quad (7)$$

between the correct frequencies $f_i$, $i = 1, 2, \ldots, n$ and their estimates $\hat{f}_{i,r}$ in $R = 200$ runs. The measurement noise samples are drawn from an i.i.d complex Gaussian random process with zero mean and variance $\sigma^2$. All the amplitudes of the complex sinusoids were considered equal to one, and the number of steps of the VP algorithm was set to 20 for all the test cases. We analysed the performance of the VP algorithm over the number of iterations and we noticed that after approximately 20 iterations the accuracy of the estimation does not change. The function `rootmusic` built in MATLAB and `varpro.m` [18] were used for obtaining the simulation results. The DTFT was evaluated numerically using the DFT with zero padding.

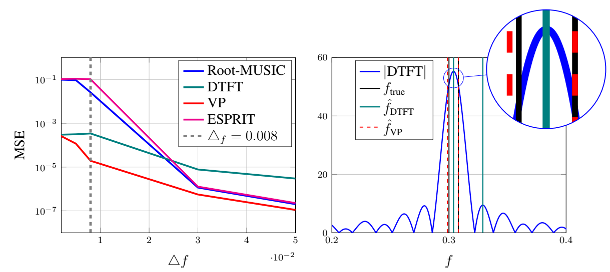
Fig. 2(a) depicts a scenario with $M = 30$ and normalized frequencies $f_1 = 0.0667, f_2 = 0.1333, f_3 = 0.20, f_4 = 0.2667, f_5 = 0.4667$. In addition to the MSE of the considered estimation methods, we also plot the Cramer-Rao lower bound (CRB) [20]. It can be noticed that for low SNRs values the performance of the DTFT and VP remains almost the same, but for high values VP considerably overcomes the DTFT. As it is known in the presence of multiple frequencies, the DTFT suffers of spectral leakage due to interference among the sinusoidal components [21] which cause it to become biased. The VP algorithm is able to counteract the effects of this phenomenon, reducing considerably the MSE and reaching the CRB. Moreover, VP outperforms Root-MUSIC and ESPRIT for all the considered SNR region. Fig. 2(b) shows a scenario with two closely spaced frequencies $f_1 = 0.405, f_2 = 0.45$. The performance gain of VP compared to Root-MUSIC and ESPRIT is most notable in this scenario, due to the fact that the super-resolution techniques (as Root-MUSIC and ESPRIT) show a lack of performance when the frequencies are placed very close to each other [22, 23]. VP also for this scenario reaches the CRB.




(a) (b)

**Fig. 2**: MSE vs SNR for a scenario with $M = 30$ (a) for five frequencies (b) two closely spaced frequencies.

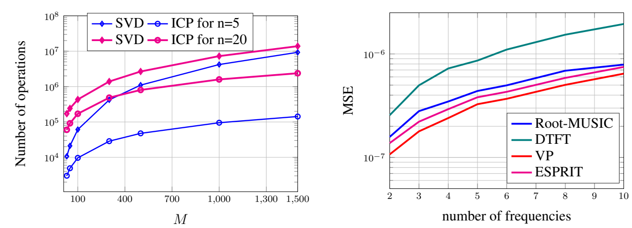
Fig. 3(a) illustrates the MSE for a scenario with $M = 30$, SNR=20dB, and two frequencies chosen such that the distance between them is equal to $\triangle_f$. When $\triangle_f$ is very small, due to the closeness of the frequencies, the DTFT cannot distinguish the two peaks. Then the search of the $n$ largest peaks on amplitude of the DTFT results in two estimated frequencies, one of them is lying between the correct values and the second one in one of the sidelobes. After applying VP optimization this effect is reduced considerably; even one of the frequency was exactly detected, see Fig. 3(b) for $\triangle_f = 0.008$ (note that they overlap). Simulation results confirm that the peaks that cannot be separated for the DTFT, can be determined after the VP optimization for values of $\triangle_f > 0.008$ and a reasonable SNR. For smaller values, due to the poor initialization, no improvements are observed in comparison with the DTFT. As expected, Root-MUSIC and ESPRIT show a lack of performance for small values of $\triangle_f$. VP overcomes Root-MUSIC and ESPRIT for all the simulated values of $\triangle_f$.




(a) MSE vs. $\triangle_f$ (b) DTFT spectrum for $\triangle_f = 0.008$

**Fig. 3**: Analysis of the impact of $\triangle_f$ in the performance.

Fig. 4(b) depicts the MSE vs. the number of frequencies for a scenario with $M = 45$ and SNR=15dB. A different set of frequencies was selected uniformly at random in the interval $(0, 0.5]$ with a minimum distance between them of $2/M$. VP shows the best performance for all the simulated cases.




(a) Operations vs. the number of measurements $M$ (b) MSE vs. the number of frequencies

**Fig. 4**: Analysis of the computational complexity, and MSE vs. the number of frequencies

## 5. CONCLUSION

In this paper, we have formulated the frequency estimation problem as an SNLLS problem. Based on this representation, we have proposed a VP optimization for finding the frequency parameters. An efficient way of calculating the exact gradient of the VP functional has been presented, with a lower computational cost than existing techniques. Simulations have shown that the proposed estimator outperforms previously reported techniques in scenarios with closely spaced frequencies and achieves more accurate results in terms of the MSE. The MATLAB implementation of the proposed method is available at the website [24].

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